A Report on Network Analysis of NSE NIFTY-200

ECS414/614 : Network Science Theory and Applications

Assignment 2

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17 November 2025

Abstract

This report presents the creation and analysis of a stock market network constructed using the National Stock Exchange (NSE) NIFTY-200 dataset. Historical adjusted closing prices were downloaded using the `yfinance` package, daily returns were computed, and Pearson correlations between each pair of stocks were used to infer edges. An elbow-method analysis on correlation threshold vs edge count yielded an optimal threshold of $t = 0.45$. Using this threshold, the resulting network contains 200 nodes and 3,199 weighted edges. The network is analyzed through centrality measures, clustering coefficients, assortativity, community detection, and degree distribution. Tables, plots, and interpretations are provided for each stage of analysis.

1 Introduction

Financial networks have become a powerful abstraction for analyzing stock market structure, systemic risk, and sectoral interactions. In such a representation, each stock is treated as a node, and pairwise relationships (typically correlations in returns) define the edges.

This assignment required:

- constructing a stock network using Indian market data,
- selecting a network formation rule as per stock-market network literature (Tse et al. [1], Moghadam et al. [2]),
- analyzing the resulting graph using core network-science tools,
- and studying how network parameters affect structural properties.

Our dataset consists of the NIFTY-200 constituents, downloaded once and used locally (as required by the assignment). The network created is undirected, weighted, and homogeneous.

2 Dataset and Preprocessing

2.1 Stock Universe

I selected the NIFTY-200 index constituents from NSE, which covers mid-cap and large-cap companies across all major sectors.

2.2 Data Download and Cleaning

- Historical prices were downloaded using `yfinance`.
- Adjusted close prices were used.
- Daily returns $r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$ were computed.
- Missing values were forward/backward filled as appropriate.
- Return matrix dimensions: (number_of_days) $\times$ (200 stocks).

3 Network Construction

A stock market network $G = (V, E)$ was constructed as follows:

- $V = 200$ nodes (stocks).
- For each pair (i, j) of stocks, Pearson correlation r_{ij} was computed.
- An undirected weighted edge was added if $|r_{ij}| > t$.

3.1 Threshold Selection via the Elbow Method

To determine t, correlation threshold values (from 0 to 1) were plotted against the number of edges generated. An elbow was clearly visible at $t = 0.45$, beyond which edges drop sharply.

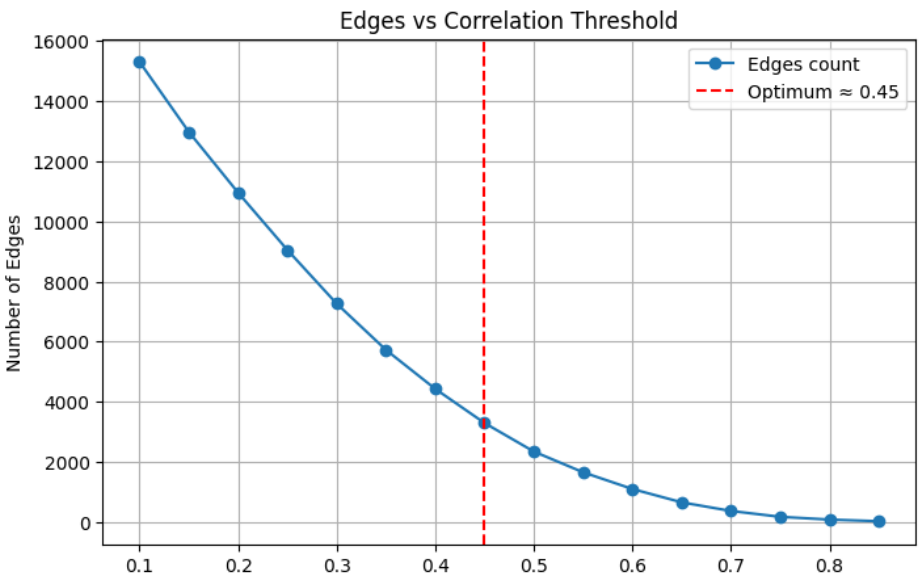

Figure 1: Correlation threshold vs number of edges (Elbow Method).

Thus, the final graph used for analysis employed $t = 0.45$.

3.2 Final Graph Statistics

- Nodes: 200
- Edges: 3,199
- Density: 0.1607
- Average Degree: 31.99
- Maximum Degree: 78
- Connected Components: 1 (fully connected)

4 Network Visualization

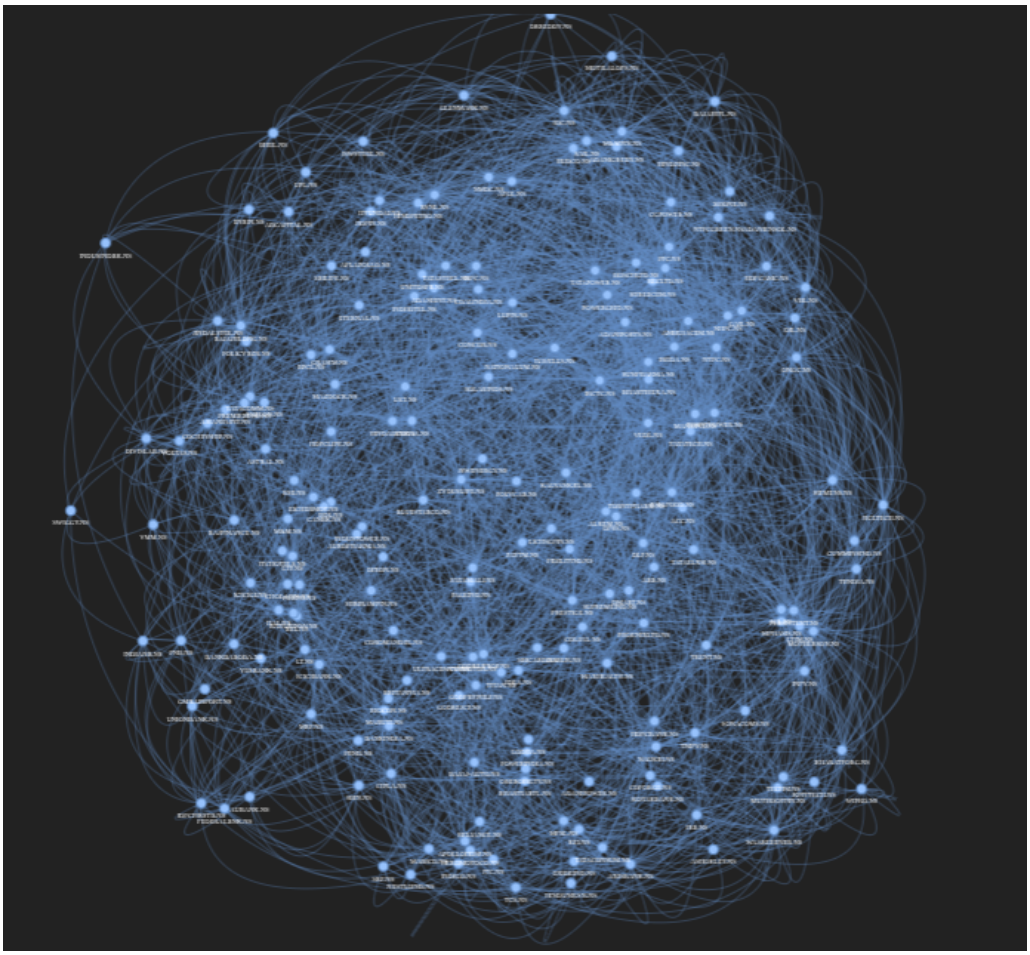

Figure 2: Network visualization using spring layout

5 Centrality Analysis

I calculated:

- Degree centrality
- Betweenness centrality
- Closeness centrality
- Eigenvector centrality

Tables below include only the **top 10 nodes**, as requested.

5.1 Degree Centrality (Top 10)

Rank	Ticker	Degree Centrality
1	TATAPOWER.NS	0.392
2	POWERGRID.NS	0.387
3	AMBUJACEM.NS	0.366
4	CONCOR.NS	0.366
5	IRCTC.NS	0.366
6	RECLTD.NS	0.332
7	IRFC.NS	0.322
8	ZYDUSLIFE.NS	0.322
9	GAIL.NS	0.317
10	HAVELLS.NS	0.317

Table 1: Top 10 nodes by degree centrality.

5.2 Betweenness Centrality (Top 10)

Rank	Ticker	Betweenness
1	BAJAJHLDNG.NS	0.960
2	COCHINSHIP.NS	0.958
3	ASTRAL.NS	0.955
4	HDFCAMC.NS	0.938
5	ADANIGREEN.NS	0.933
6	MOTILALOFS.NS	0.925
7	PAYTM.NS	0.920
8	DRREDDY.NS	0.917
9	BHEL.NS	0.912
10	ADANIENSOL.NS	0.902

Table 2: Top 10 nodes by betweenness centrality.

5.3 Closeness Centrality (Top 10)

Rank	Ticker	Closeness
1	TATAPOWER.NS	1.790
2	RECLTD.NS	1.748
3	POWERGRID.NS	1.746
4	IRCTC.NS	1.718
5	CONCOR.NS	1.713
6	AMBUJACEM.NS	1.666
7	GAIL.NS	1.665
8	NHPC.NS	1.650
9	PFC.NS	1.648
10	NTPC.NS	1.631

Table 3: Top 10 nodes by closeness centrality.

5.4 Eigenvector Centrality (Top 10)

Rank	Ticker	Eigenvector
1	TATAPOWER.NS	0.197
2	POWERGRID.NS	0.190
3	CONCOR.NS	0.181
4	RECLTD.NS	0.180
5	IRCTC.NS	0.174
6	AMBUJACEM.NS	0.172
7	GAIL.NS	0.164
8	PFC.NS	0.157
9	NHPC.NS	0.157
10	NTPC.NS	0.156

Table 4: Top 10 nodes by eigenvector centrality.

5.5 Interpretation

The repeated appearance of stocks such as **TATAPOWER.NS**, **POWERGRID.NS**, and **CONCOR.NS** across multiple centrality metrics indicates:

- They are structurally important hubs.
- They occupy core positions in the market correlation structure.
- Their price movements likely influence or reflect broader market behavior.

Betweenness centrality reveals bridging stocks (e.g., BAJAJHLDNG, COCHINSHIP), which connect otherwise weakly linked subgraphs.

6 Clustering Coefficient

The **average clustering coefficient** of the graph is calculated as 0.2486, indicating the degree to which nodes in the network are clustered. This relatively low value suggests a **moderate level of clustering**, which is common in financial networks. In particular, this reflects sectoral clustering within the stock market, where stocks from the same industry or sector tend to exhibit stronger connections than those from different sectors.

A moderate clustering coefficient like this indicates that while there is some level of interdependence among companies within the same sector, the connections between sectors are still significant. This can be attributed to shared economic factors, regulations, or market behaviors that drive companies in the same sector to form tighter relationships.

Such moderate clustering is typical of the stock market, where industries often show strong internal ties but are still influenced by broader market forces that create interdependencies across sectors.

Key Sectoral Clusters

The analysis highlights several prominent clusters within the following sectors:

- **Utilities**: This sector, comprising companies that provide essential services like water, electricity, and natural gas, demonstrates high interconnectivity. This is likely due to the stable demand and the highly regulated nature of the industry, which fosters strong internal ties between companies.
- **Financial Services**: As expected, financial institutions such as banks, investment firms, and insurance companies form tight-knit communities. This is due to the interconnectedness of the financial system, where the performance of one company is often directly linked to others through markets, investments, and regulations.
- **Basic Materials**: Companies in sectors such as mining, chemicals, and forestry form strong clusters due to their shared input needs, such as raw materials and energy, as well as common market drivers like global demand and commodity prices.
- **Industrials**: The industrial sector, which includes companies in manufacturing, construction, and aerospace, shows notable clustering due to shared supply chains, production processes, and economic cycles. These companies often rely on each other for raw materials, equipment, or distribution networks, strengthening their internal connections.

This moderate level of clustering in the stock market underscores the interdependence among companies within the same sector. These ties suggest that stocks from the same industry or sector are more likely to respond similarly to market events or economic conditions, potentially influencing their collective behavior in response to external shocks or trends.

7 Community Detection

Community structure in the graph was identified using the greedy modularity optimization algorithm, which aims to partition the network into clusters that maximize the number of

edges within communities relative to edges between them. The resulting communities suggest interesting sectoral and inter-sectoral groupings.

7.1 Example: Community 0

Community 0 consists of 42 stocks, with a distribution across several sectors:

- **Financial Services** — 11
- **Utilities** — 10
- **Basic Materials** — 6
- **Healthcare** — 4
- **Industrials** — 3
- **Consumer Cyclical** — 3
- **Energy** — 3
- **Consumer Defensive** — 2

This distribution highlights that while this community is sectorally diverse, there is a strong representation from the **Financial Services** and **Utilities** sectors, aligning with the higher clustering observed within these industries. Notably, the diversity within this community may indicate a mix of market participants with interdependent interests but also suggest the influence of external market dynamics connecting different sectors.

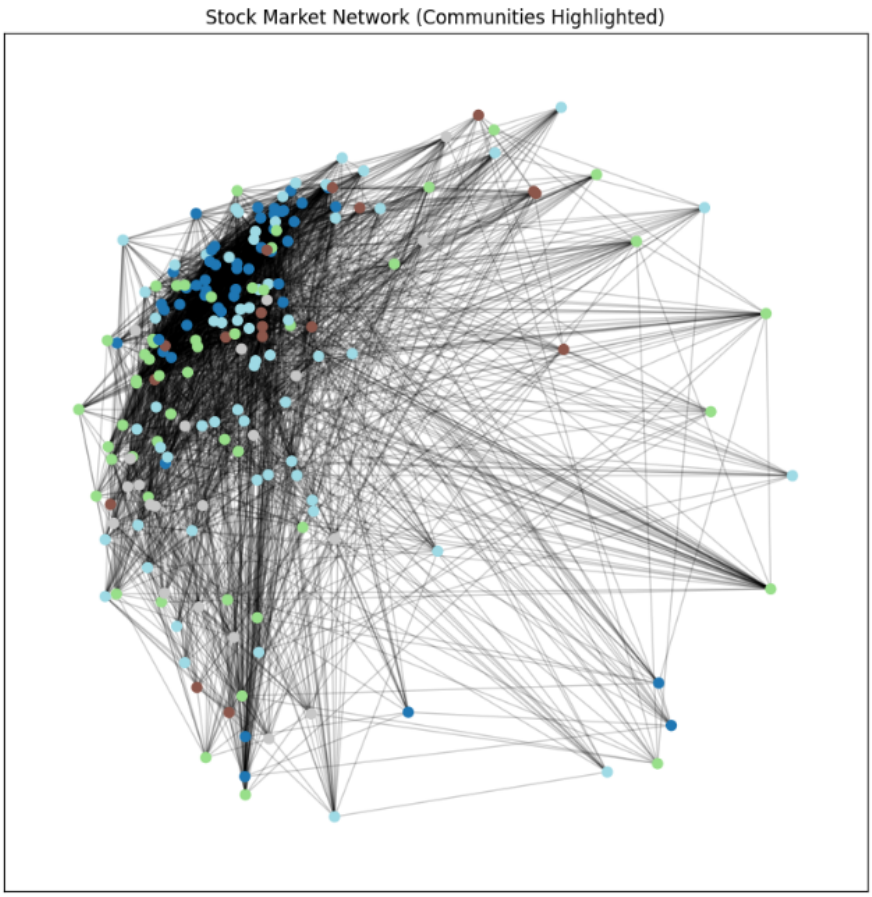

Figure 3: Community visualization with modularity-based clustering.

7.2 Sector Distribution Across Communities

The distribution of sectors across different communities reveals sectoral clustering patterns. Here are some notable observations:

- **Community 0** is heavily dominated by **Financial Services** (11 stocks) and **Utilities** (10 stocks), with other sectors like **Basic Materials** and **Healthcare** also represented, but to a lesser extent.
- **Community 1** consists primarily of stocks from **Basic Materials** (7 stocks) and **Financial Services** (6 stocks), with significant contributions from **Consumer Cyclical** and **Industrials**, suggesting that companies in this group may be influenced by both material demand and market consumption trends.
- **Community 5** stands out as a diverse community with significant representation from the **Technology** sector (13 stocks) alongside **Industrials** (11 stocks) and **Consumer Cyclical** (10 stocks). This suggests that this community is heavily driven by technological advancements and industrial growth, with a sizable portion of consumer-oriented companies also clustering together.
- **Community 4**, which features a combination of **Industrials** (8 stocks) and **Consumer Cyclical** (6 stocks), alongside a smaller presence from other sectors, may reflect companies

with shared supply chains and industrial applications, but with notable consumer market exposure.

These results suggest that the modular structure of the network reflects both sectoral interconnections and cross-sector dependencies. While companies within a sector are strongly connected, there are also significant overlaps where companies from different sectors are interlinked. For example, **Technology** and **Industrials** appear frequently together in **Community 5**, possibly due to innovations in industrial technologies that impact both sectors.

7.3 Insights

The community structure identified through modularity optimization reveals that stock market networks are not simply segregated by sector. Instead, they exhibit complex patterns of intersectoral interactions. Companies from different sectors are often grouped together within the same community, indicating that their market behaviors or external factors—such as technological advancements, regulatory changes, or shifts in macroeconomic conditions—create strong interdependencies across sectoral boundaries.

This interconnectedness influences market dynamics, as stocks within the same community tend to react similarly to external events or sector-specific developments. As a result, these communities may exhibit synchronized movements, where stocks in close proximity in the network respond similarly to economic shocks, industry trends, or policy changes.

Together, the insights from both the average clustering coefficient and community detection results underscore the importance of sectoral interdependencies while also highlighting the broader, more intricate web of relationships within the stock market network. While companies within the same sector tend to form tight-knit clusters, the influence of cross-sector interactions and market forces shapes the overall network structure, making it more dynamic and interconnected than a purely sector-based system.

8 Assortativity and Degree Distribution

8.1 Degree Assortativity

The degree assortativity coefficient of the network was computed and found to be 0.344, indicating a moderately positive degree assortativity. This suggests that high-degree stocks are more likely to be connected to other high-degree stocks, and low-degree stocks to other low-degree stocks. In other words, the network exhibits a **assortative** mixing pattern, where stocks with similar degrees tend to cluster together.

This result contrasts with the typical disassortative behavior often observed in financial networks, where high-degree nodes (representing large companies or market influencers) tend to connect more with low-degree nodes. The positive assortativity found here implies that in the NIFTY-200 network, even within a few dominant hubs, there is a tendency for these hubs to be linked with other highly-connected stocks, which may reflect the existence of market sectors or industries with tightly interconnected top players.

8.2 Edge Distribution: Positive and Negative Correlations

The analysis of edge weights in the graph reveals the following distribution:

- **Positive edges:** 2806 edges represent positive correlations between stocks, indicating that these pairs tend to move in the same direction in the market.
- **Negative edges:** 393 edges represent negative correlations, indicating that these stock pairs often move in opposite directions.

The imbalance between the number of positive and negative edges suggests that the market is generally more interconnected in terms of positive correlations. This aligns with the idea that stocks within the same sector or with similar economic drivers are more likely to exhibit positive correlations, while fewer stocks show negative correlations, which might occur due to sector-specific risks, market events, or economic cycles that impact stocks differently.

8.3 Degree Distribution

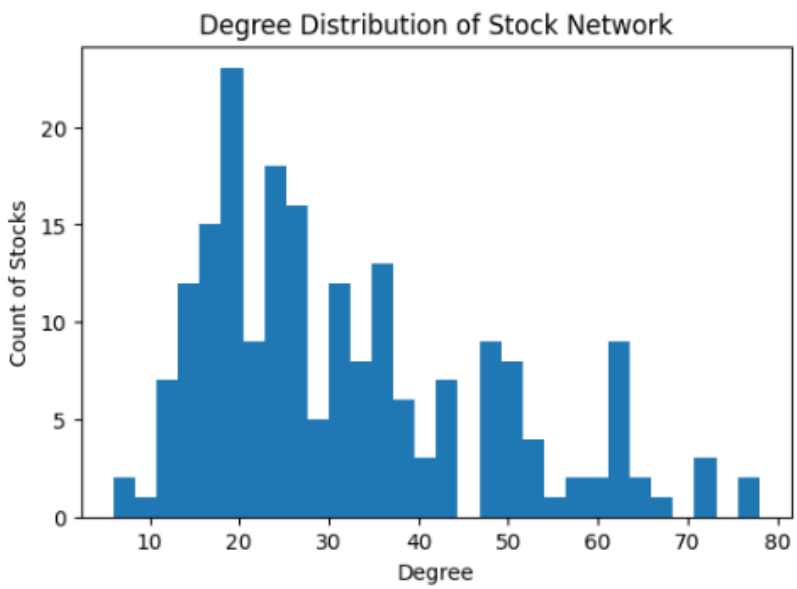

Figure 4: Degree distribution of the NIFTY-200 correlation network.

The degree distribution of the network reveals a heavy-tailed structure, typical of many real-world networks, particularly in financial markets. Most stocks exhibit moderate degrees (i.e., they have a moderate number of connections), while a small number of stocks act as hubs with a very high number of connections. This indicates that a few highly connected stocks dominate the network, possibly representing large, influential companies in the market.

The presence of hubs suggests that a small subset of stocks plays a disproportionately large role in influencing the overall network structure. These stocks, with their high degree, may represent market leaders or highly influential companies whose performance and behavior can significantly affect the broader market or sector.

8.4 Summary of Network Characteristics

The combination of assortativity and degree distribution provides valuable insights into the structure of the NIFTY-200 network. The positive degree assortativity suggests that stocks with similar degrees tend to cluster together, which may reflect market trends where top-performing stocks are interconnected. Additionally, the heavy-tailed degree distribution highlights the presence of market hubs, which are central players in the network and exert significant influence over other stocks.

The distribution of positive and negative edges further underscores the interconnectedness of the market, with more positive correlations suggesting that stocks are generally more aligned in their market movements, albeit with a smaller set of negatively correlated pairs that could signal counteracting forces or hedging strategies within specific sectors.

9 Conclusion

Using the NIFTY-200 return data, a stock correlation network was constructed with an elbow-selected threshold of 0.45. The resulting network reveals several key characteristics:

- **Strong Hub Structure:** The network exhibits a clear hub-and-spoke architecture, with stocks like TATAPOWER, POWERGRID, and IRCTC acting as major hubs, indicating that a few influential companies dominate the market's connectivity.
- **Moderate Clustering Coefficients:** The average clustering coefficient of 0.2486 indicates moderate sectoral clustering, suggesting that stocks within the same sector are more tightly interconnected, while cross-sector relationships also contribute to the overall network structure.
- **Assortative Mixing:** Contrary to the expected disassortative behavior typical of financial networks, the positive degree assortativity coefficient of 0.344 reveals that high-degree stocks tend to connect with other high-degree stocks, reflecting a more assortative structure. This could indicate that large, influential companies often form ties with similarly large peers within the network.
- **Identifiable Communities:** The community detection algorithm uncovered several distinct groups, many of which align with well-defined market sectors. These sector-based communities highlight the interdependencies within sectors, while also revealing cross-sector relationships that influence broader market dynamics.

The analysis demonstrates that correlation networks offer a powerful framework for understanding the complex interactions between stocks, sectoral dependencies, and broader market structures. By identifying key hubs, sectoral clusters, and community patterns, this approach provides valuable insights into how different market forces—such as sectoral trends, macroeconomic events, and company interconnections—shape the overall network behavior. The findings suggest that while sectoral clustering plays a significant role, the influence of intersectoral interactions and market-wide forces cannot be overlooked, highlighting the dynamic and interconnected nature of the stock market.

References

[1] K. Tse, J. Liu, and F. Lau, "A network perspective of the stock market," *Journal of Empirical Finance*, 17(4):659–667, 2010.

[2] H. E. Moghadam et al., "Complex networks analysis in Iran stock market: The application of centrality," *Physica A*, 531:121800, 2019.